

Astro 142

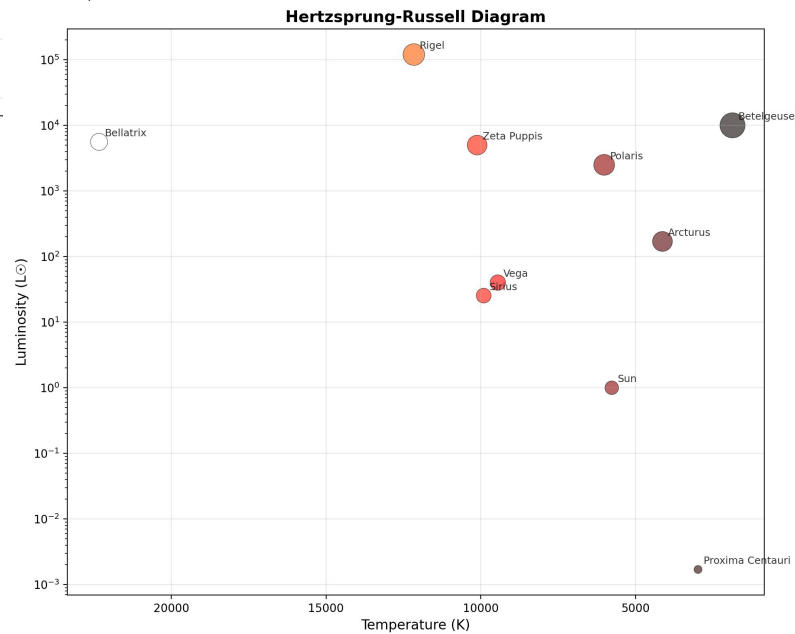
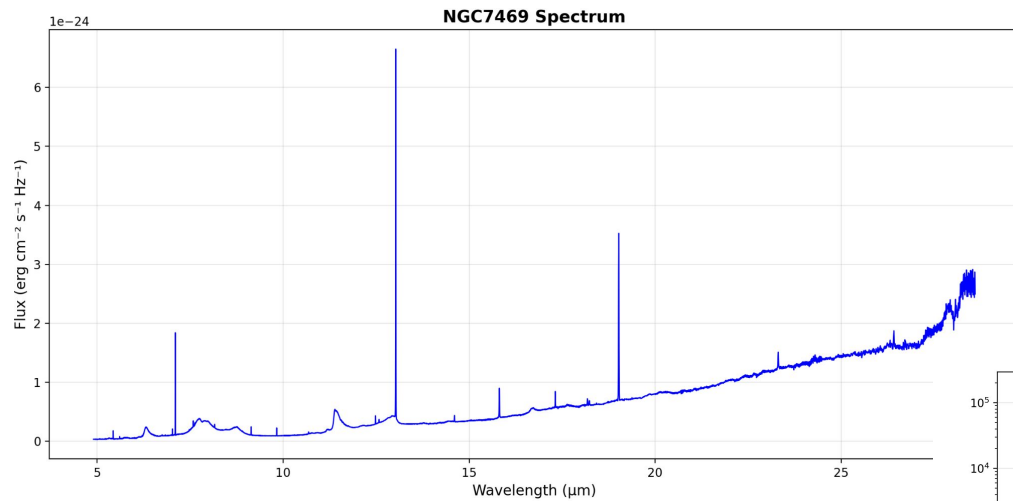
Discussion 11

2025-11-04
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Discussion Activity

Discussion 11 Activity

Download the python script and data files. Follow the instructions in the script and attach screenshots to the next slides. Upload a pdf of your slides when you are done.



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9
10 # Task 1
11 # Load the NGC7469 spectrum and convert flux units
12 hdul = fits.open('NGC7469_SingleExt_r0.4as_cube.fits')
13
14 flux_jy = hdul['FLUX_ST'].data
15 wavelength_micron = hdul['WAVE'].data
16
17 flux_jy_units = flux_jy << u.Jy
18 flux_cgs = flux_jy_units.to(u.erg / u.cm**2 / u.s / u.Hz)
19
20 plt.figure(figsize=(12, 6))
21 plt.plot(wavelength_micron, flux_cgs.value, 'b-', linewidth=1)
22 plt.xlabel('Wavelength (μm)', fontsize=12)
23 plt.ylabel('Flux (erg cm-2 s-1 Hz-1)', fontsize=12)
24 plt.title('NGC7469 Spectrum', fontsize=14, fontweight='bold')
25 plt.grid(True, alpha=0.3)
26 plt.tight_layout()
27 plt.savefig('ngc7469_spectrum.png', dpi=300)
28 print("\nSpectrum plot saved as 'ngc7469_spectrum.png'")
29 plt.show()
30
31 hdul.close()
32
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33 # Task 2
34 star_table = Table.read('star_list.csv', format='ascii.csv')
35
36 print("\nOriginal star table:")
37 print(star_table)
38
39 coords = SkyCoord(ra=star_table['RA (deg)'] << u.deg,
40                  dec=star_table['DEC (deg)'] << u.deg,
41                  frame='icrs')
42
43 ra_hms = coords.ra.to_string(unit=u.hour, sep=':', precision=2)
44 dec_dms = coords.dec.to_string(unit=u.degree, sep=':', precision=2)
45
46 star_table['RA (HMS)'] = ra_hms
47 star_table['DEC (DMS)'] = dec_dms
48
49
50 sigma_sb = 5.670374419e-5
51 L_sun = 3.828e33
52 R_sun = 6.957e10
53
54 luminosity_erg_s = star_table['Luminosity (Lsun)'] * L_sun
55 radius_cm = star_table['Radius (Rsun)'] * R_sun
56
57 temperature_K = (luminosity_erg_s / (4 * np.pi * radius_cm**2 * sigma_sb))**0.25
58
59 star_table['Temperature (K)'] = temperature_K
60
61
62 # Add two more stars
63 new_stars = Table({
64     'Star': ['Proxima Centauri', 'Polaris'],
65     'RA (deg)': [217.429, 37.955],
66     'DEC (deg)': [-62.679, 89.264],
67     'Luminosity (Lsun)': [0.0017, 2500.0],
68     'Radius (Rsun)': [0.154, 46.0],
69     'Mass (Msun)': [0.122, 5.4],
70     'Distance (pc)': [1.301, 133.0]
71 })
72
73 new_coords = SkyCoord(ra=new_stars['RA (deg)'] << u.deg,
74                      dec=new_stars['DEC (deg)'] << u.deg,
75                      frame='icrs')
76
77 new_stars['RA (HMS)'] = new_coords.ra.to_string(unit=u.hour, sep=':', precision=2)
78 new_stars['DEC (DMS)'] = new_coords.dec.to_string(unit=u.degree, sep=':', precision=2)
79

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80 new_luminosity = new_stars['Luminosity (Lsun)'] * L_sun
81 new_radius = new_stars['Radius (Rsun)'] * R_sun
82 new_temperature = (new_luminosity / (4 * np.pi * new_radius**2 * sigma_sb))**0.25
83
84 new_stars['Temperature (K)'] = new_temperature
85 complete_table = vstack([star_table, new_stars])
86
87 # Save the updated table
88 complete_table.write('star_list_updated.csv', format='ascii.csv', overwrite=True)
89
90
91 plt.figure(figsize=(10, 8))
92 marker_sizes = 50 + 450 * (np.log10(complete_table['Radius (Rsun)']) - np.log10(complete_table['Radius (Rsun)'].min())) / \
93     (np.log10(complete_table['Radius (Rsun)'].max()) - np.log10(complete_table['Radius (Rsun)'].min()))
94
95 plt.scatter(complete_table['Temperature (K)'], complete_table['Luminosity (Lsun)'],
96            s=marker_sizes, alpha=0.6,
97            c=complete_table['Temperature (K)'], cmap='hot', edgecolors='black', linewidth=0.5)
98 plt.xlabel('Temperature (K)', fontsize=12)
99 plt.ylabel('Luminosity (Lo)', fontsize=12)
100 plt.yscale('log')
101 plt.gca().invert_yaxis()
102 plt.title('Hertzsprung-Russell Diagram', fontsize=14, fontweight='bold')
103 plt.grid(True, alpha=0.3)
104 for i, star in enumerate(complete_table['Star']):
105     plt.annotate(star, (complete_table['Temperature (K)'][i],
106                       complete_table['Luminosity (Lsun)'][i]),
107                fontsize=9, alpha=0.8, xytext=(5, 5), textcoords='offset points')
108
109 plt.tight_layout()
110 plt.savefig('hr_diagram.png', dpi=300)
111 plt.show()

```